

**UNITED STATES INTERNATIONAL TRADE COMMISSION
WASHINGTON, D.C.**

**Before the Honorable Clark S. Cheney
Chief Administrative Law Judge**

In the Matter of

**CERTAIN SEMICONDUCTOR DEVICES,
PRODUCTS CONTAINING SAME, AND
COMPONENTS THEREOF**

Investigation No. 337-TA-1500

**COMPLAINANT GLOBALFOUNDRIES U.S. INC.'S IDENTIFICATION OF EXPERT
WITNESSES FOR CLAIM CONSTRUCTION**

Pursuant to the Procedural Schedule (Order No. 6), Complainant GlobalFoundries U.S. Inc. (“GlobalFoundries”) provides this disclosure of expert witnesses who may offer opinion or testimony for the *Markman* hearing. Attached hereto as **Exhibits A-C** are copies of the experts’ curricula vitae. GlobalFoundries reserves the right to supplement or amend this disclosure as appropriate.

1. David Kuan-Yu Liu, Ph.D.

Dr. David Kuan-Yu Liu received a Bachelor of Science in Electrical Engineering from the University of California, Berkeley, in 1983. He received a Master of Science and Ph.D. in Electrical Engineering from Stanford University in 1985 and 1989, respectively. From 1989 to 1992, he was a member of technical staff at Texas Instruments, Inc. (“TI”). At TI, his job responsibilities included process integration, device modeling, high-voltage complementary metal oxide semiconductor (CMOS) process integration, and investigating novel source-side injection mechanisms for Flash EPROM channel hot-electron programming. From 1992 to 1995, he was a member of the technical staff at Advanced Micro Devices, Inc. (“AMD”) where he was a key contributor in optimizing Flash cell and periphery devices in AMD’s CMOS-based 0.5µm and

0.35 μ m Flash EPROM technology. His job responsibilities at AMD also included process integration, device modeling, and development of triple-well process technology to accommodate x-decoder transistors and high-voltage transistors for negative gate erase operations. While at AMD, he was awarded a Spotlight Award for developing a method of manufacturing a self-aligned source (SAS) etch for a NOR flash memory. He spent the next five years in managerial and director roles at several California-based semiconductor companies, where he was responsible for increasing yield and for leading teams of engineers working to develop next generation memory devices. In 2000, he co-founded Progressant Technologies in Fremont, California. Progressant Technologies developed IP for negative differential resistance transistor technology and was eventually acquired by Synopsys, Inc. From 2000 to 2004, I was a Senior Manager at Xilinx, Incorporated, where he was responsible for developing nonvolatile memory process technology for flash and CPLD product applications, as well as advanced CMOS process technology (specifically 75nm CMOS technology node, a half node version between 90nm and 65nm). From 2004 to 2007, he was a Senior Scientist at Maxim Integrated Products where he was responsible for developing Embedded Non-volatile Memory process technology for Power Management product applications. Since 2007, he has served as a technical consultant, providing expert advice regarding semiconductor memory technologies, including CMOS process technology, and semiconductor device physics, as set forth in his curriculum vitae, attached hereto as **Exhibit A**. For the purposes of claim construction, Dr. Liu may be called to testify regarding the technology related to the Asserted Patents, how a person of ordinary skill in the art would or would not understand the scope of the purported invention, and how a person of ordinary skill in the art would interpret the claims in the patent. Dr. Liu may also rebut any expert witness testimony of Respondents' expert(s) as appropriate.

2. Jack C. Lee, Ph.D.

Dr. Jack C. Lee is an expert in the field of semiconductor design and process technology and have forty years of first-hand experience as a researcher, educator, and consultant in this field. Dr. Lee has a B.S. degree in Electrical Engineering, with highest honors, in 1980, and an M.S. degree in Electrical Engineering in 1981, both from University of California, Los Angeles. Dr. Lee received a Ph.D. degree in Electrical Engineering in 1988 from University of California, Berkeley (“UC Berkeley”). After receiving his Ph.D. in August 1988, Dr. Lee joined the faculty at The University of Texas at Austin (“UT Austin”). As a faculty member, Dr. Lee has taught numerous courses in semiconductor device fabrication and design, at both the undergraduate and graduate levels. He has supervised over 40 students who received their doctoral degrees under his guidance. Since September 2000, he has held the Cullen Trust for Higher Education Endowed Professor in Engineering in the Department of Electrical and Computer Engineering at UT Austin. In September 2023, he was promoted to the Professor Emeritus level. His current research and teaching interests encompass semiconductor devices including 3D structures such as FinFETs or tri-gate MOSFETs, high-K gate dielectrics and gate electrodes, fabrication processes, electrical and material characterization, reliability and modeling, packaging technologies, dielectric and passivation processes, display technologies, photovoltaic and solar cell technologies, semiconductor memory applications, and alternative channel materials, as set forth in his curriculum vitae, attached hereto as **Exhibit B**. For the purposes of claim construction, Dr. Lee may be called to testify regarding the technology related to the Asserted Patents, how a person of ordinary skill in the art would or would not understand the scope of the purported invention, and how a person of ordinary skill in the art would interpret the claims in the patent. Dr. Lee may also rebut any expert witness testimony of Respondents’ expert(s) as appropriate.

3. Massoud Pedram, Ph. D.

Dr. Massoud Pedram is a Professor of Electrical Engineering and the Charles Lee Powell Chair in Electrical Engineering and Computer Science in the Viterbi School of Engineering at the University of Southern California (“USC”). He is an expert on Very Large Scale Integration (“VLSI”) design including parallel computing, distributed processing, high-speed design, power efficient design, digital integrated circuits, and semiconductor memory. He also has expertise in neural networks and graphic processing units (“GPUs”), as well as system-on-chip designs targeting applications ranging from high-performance computing to low-power embedded computing. His current research interests are in, among others, computer-aided design methodologies and tools for CMOS circuits and systems, energy-efficient computing, computer architecture and hardware, parallel and distributed computing, neural network acceleration, and integrated circuit and system design including system-on-chip design. He earned his B.S. in Electrical Engineering from the California Institute of Technology in 1986, and his M.S. and Ph.D. in Electrical Engineering and Computer Sciences from the University of California at Berkeley in 1989 and 1991, respectively, as set forth in his curriculum vitae, attached hereto as **Exhibit C**. For the purposes of claim construction, Dr. Pedram may be called to testify regarding the technology related to the Asserted Patents, how a person of ordinary skill in the art would or would not understand the scope of the purported invention, and how a person of ordinary skill in the art would interpret the claims in the patent. Dr. Pedram may also rebut any expert witness testimony of Respondents’ expert(s) as appropriate.

Dated: August 17, 2026

Respectfully submitted,

/s/ Bert C. Reiser

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Counsel for Complainant

GlobalFoundries U.S. Inc.

CERTIFICATE OF SERVICE

It is hereby certified that copies of the foregoing **COMPLAINANT GLOBALFOUNDRIES U.S. INC.'S IDENTIFICATION OF EXPERTS FOR CLAIM CONSTRUCTION** were served on August 17, 2026 as follows:

The Honorable Lisa R. Barton Secretary to the Commission U.S. International Trade Commission 500 E Street, SW, Room 112 Washington, DC 20436	BY EDIS
The Honorable Clark S. Cheney Chief Administrative Law Judge U.S. International Trade Commission 500 E Street, SW, Room 317 Washington, DC 20436	By Email and Box: Cheney1500@usitc.gov
Stephen R. Smith COOLEY LLP 1299 Pennsylvania Ave., N.W. Suite 700 Washington, D.C. 20004 <i>Counsel for Respondents Tower Semiconductor Ltd., Tower Partners Semiconductor, Ltd., Tower Semiconductor Italy S.r.l., Tower US Holdings, Inc., Tower Semiconductor San Antonio, Inc., Tower Semiconductor NPB Holdings, Inc., Tower Semiconductor Newport Beach, Inc., and Newport Fab LLC</i>	By Email: TowerITC@cooley.com

/s/ Erika J. Weinstein

Erika J. Weinstein

CLEARY GOTTlieb STEEN & HAMILTON LLP

EXHIBIT A

DAVID KUAN-YU LIU, Ph.D.

davidkyliu@comcast.net

PROFESSIONAL PROFILE

- Technical expert with over 20 years of industrial experience in the development of advanced CMOS devices and process technology, with special emphasis on non-volatile semiconductor memory devices and process technology, and high-voltage semiconductor devices, with additional 18 years of experience serving as expert consultant.
 - Named inventor on 99 US patents.
- Consultant and expert witness for past 18 years in semiconductor intellectual property evaluation and patent litigation, with deposition experience and successful ITC trial testifying experience.
- Successful track record in transitioning products from development phase to high-volume manufacturing phase, both as an individual contributor and as manager/director.
- Co-founder of an LLC in the development and licensing of intellectual property with emphasis on embedded flash memory technology.
- Co-founder of a semiconductor memory start-up company (Progressant Technologies, Inc.); participated in its successful sale to Synopsys, Inc.
- Fluent in Mandarin Chinese (native language) and English; over 10 years experience as real-time translator of church sermons and proficient in written translation.

EDUCATION

Stanford University, Palo Alto, California

Ph.D., Electrical Engineering (1989)

Thesis: Physics and Technology of Novel Conductivity-Modulated Power MOSFET's

M.S., Electrical Engineering (1985)

Coursework in integrated-circuit technology, semiconductor device physics, computer modeling of semiconductor devices and fabrication processes, and solid-state physics

University of California, Berkeley, California

B.S., Electrical Engineering (1983)

Certificate of Distinction (awarded annually to the top EECS graduate)

EXPERT WITNESS WORK - CASE EXPERIENCE:

Case: **Advanced Memory Technologies, Inc. v. Micron Technology, Inc.**
Jurisdiction: Civil Action No. 1:25-cv-01036 (W.D. Tex.)
Work Product: Expert consulting work relating to Booster Circuit. Patent analysis and prior art review toward preparation of EPR petition support and draft of declaration.
Nature of Case: EPR.
Counsel: Retained by Fish & Richardson (Contact: Tim Riffe)
Date: 11/25 – Present

Case: **Apple v. WeCrevention, Inc.**
Jurisdiction: Post-Grant Patent Matters Adverse to WeCrevention, Inc.
Work Product: Expert consulting work relating to High Speed Chip Memory Module. Patent analysis and prior art review toward preparation of IPR petition support and draft of declaration.
Nature of Case: IPR.
Counsel: Retained by Fish & Richardson (Contact: Dr. Hyun Jin In)
Date: 11/25 – Present

Case: **Monterey Research LLC v. Renesas Electronics Corp.**
Jurisdiction: Civil Action No. 2:24-cv-00238-JRG (E.D. Tex.)
Work Product: Expert consulting work relating to Charge Trapping Nonvolatile Memory. Patent analysis and prior art review toward preparation of IPR petition support and draft of declaration.
Nature of Case: Civil action and IPR.
Counsel: Retained by Morrison Foerster (Contact: Yuka Teraguchi)
Date: 03/25 – Present

Case: **Palisade Techs., LLP v. Micron Tech., Inc.**
Jurisdiction: Civil Action N.D. Cal, No. 7:24-cv-00262 (W.D. Tex.)
Work Product: Expert consulting work relating to 3D NAND Flash Memory Technology. Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 2 patents.
Nature of Case: Civil action and IPR.
Counsel: Retained by O'Melveny & Myers LLP (Contact: John Kappos)
Date: 12/24 – 01/26

Case: **Yangtze Memory Technologies Company, Ltd. vs. Micron Technology, Inc.**
Jurisdiction: Civil Action N.D. Cal, No. 3:223-cv-05792-RFL
Easter District of Texas
Work Product: Expert consulting work relating to 3D NAND Flash Memory Technology. Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 4 patents. ***Deposed on 3 petitions that have been instituted.***
Nature of Case: Civil action and IPR.
Counsel: Retained by Orrick, Herrington & Sutcliffe (Contact: Jason Lang)
Date: 03/24 – Present

Case: **Force MOS Technology Co., Ltd. vs. Inergy Technology, Inc.**
Jurisdiction: Civil Action E.D. TX, No. 2:22-cv-00460-JRG
Easter District of Texas
Work Product: Expert consulting work relating to Vertical Trench DMOS process technology. Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 2 patents. Declaration for claim construction on 3 patents. ***Deposed on Claim Construction Declaration, and deposed on declarations for IPR petitions, and deposed on reply declaration.*** Continuing work on claim construction and technical consulting.
Nature of Case: Civil action and IPR.
Counsel: Retained by Baker McKenzie (Contact: McKenzie Martin)
Date: 10/23 – 10/25

Case: **GlobalFoundries U.S. vs. International Business Machines**
Jurisdiction: 03348-KMK-AEK (S.D.N.Y. 2023)
Work Product: Expert consulting work relating to trade secrets involving advanced CMOS process and device technology (Nanosheet technology). Reviewed evidence and summarized findings.
Nature of Case: Civil action.
Counsel: Retained by Latham & Watkins (Contact: Sami Al-Marzoog)
Date: 10/23 – 2/25

Case: **Element Capital. vs. Beijing BOE Display Technology Co., Ltd.**
Jurisdiction: Civil Action No. 2:22-cv-00118-JRG
Eastern District of Texas
Work Product: Expert consulting work relating to OLED technology. Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration on 3 patents.
Nature of Case: Civil action.
Counsel: Retained by Fish & Richardson (Contact: Linhong Zhang)
Date: 4/23 – 12/23

Case: **Sonrai Memory Ltd. vs. Micron Technology, Inc.,**
Jurisdiction: Civil Action No. 6:22-cv-000855-FB
Western District of Texas
Work Product: Patent analysis and prior art review toward preparation of IPR
petition material and draft of declaration resulting in institution of
the petition and ***deposed on supplemental declaration.***
Nature of Case: Civil action.
Counsel: Retained by Fish & Richardson (Contact: Tim Riffe)
Date: 12/22 – 7/24

Case: **Daedalus Prime LLC vs. Samsung Electronics Co.,**
Jurisdiction: Civil Action E.D. TX, Marshall Div., No. 2:22-cv-353
Eastern District of Texas, Marshall Div., and
Investigation No. 337-TA-1336
Work Product: Expert consulting work relating to FINFET process technology.
Patent analysis and prior art review toward preparation of IPR
petition material and draft of declaration on 2 patents. Source code
database viewing (GDS layout and process flow documents) for
supporting of non-infringement position.
Nature of Case: Civil action and ITC Investigation on Certain Semiconductor Devices,
Mobile Devices Containing the Same and Components Thereof..
Counsel: Retained by Fish & Richardson (Contact: Kevin Su)
Date: 10/22 – 4/23

Case: **Netlist, Inc. vs. Samsung Electronics Co. Ltd. et al.,**
Jurisdiction: Civil Action No. 2:21-cv-00463
Eastern District of Texas
Work Product: Expert consulting work relating to memory modules.
Nature of Case: Civil action.
Counsel: Retained by Fish & Richardson (Contact: Tony Nguyen)
Date: 8/22 – 12/22

Case: **Vervain, LLC vs. Micron Technology & related IPR Petition**
Jurisdiction: Action No. 6:21-cv-00487-ADA
Western District of Texas
Work Product: Patent analysis and prior art review toward preparation of IPR
petition material and draft of declaration on 4 patents, all 4 petitions
instituted, and ***deposed on both initial declaration and reply
declaration on all 4 patents.***
Counsel: Retained by Orrick, Herrington & Sutcliffe LLP (Contact: Jared
Bobrow)
Date: 8/21 – 2/23

Case:	Monterey Research vs. Qualcomm PTAB Matter
Work Product:	Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel:	Retained by Jones Day (Contact: David Cochran)
Date:	12/21 – 2/22
Case:	<i>Semiconductor Energy Laboratory Co., OLED Patent Matters</i>
Jurisdiction:	Action No. 8:21-cv-00554-JAK-ADS Central District of California
Work Product:	Expert for infringement on 3 patents relating to OLED driver circuit and layout, responsibilities include source code viewing, GDS layers viewing, and claim charts review.
Nature of Case:	Civil action.
Counsel:	Retained by Fish & Richardson (Contact: Bryan Basso)
Date:	9/21 – 4/22
Case:	<i>Apple Inc. IPR Petition for U.S. Patents 6,622,108 & 6,807,505</i>
Work Product:	Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel:	Retained by Erise IP (Contact: Paul Hart)
Date:	8/21 – 9/21
Case:	<i>Texas Instruments IPR Petition</i>
Work Product:	Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.
Counsel:	Retained by Fish & Richardson (Contact: Josh Griswold)
Date:	9/20 – 4/21
Case:	<i>Micron Technology vs. UMC/JinHua</i>
Jurisdiction:	Case No. 4:17-CV-06932-MMC United States District Court for the Northern District of California
Work Product:	Testifying expert, trade secret documents review, and declaration submission.
Nature of Case:	Trade secret contention of Micron confidential documents.
Counsel:	Retained by Jones Day (Contact: Randy Kay)
Date:	3/17 – 12/23
Case:	<i>SK Hynix vs. BitMicro</i>
Jurisdiction:	Investigation No. 337-TA-1097 United States International Trade Commission
Work Product:	Testifying expert for non-infringement on 2 patents, issued expert rebuttal report, <i>deposition taken</i> , issued witness statement.
Nature of Case:	ITC infringement case concerning module stacking of SDRAM.
Counsel:	Retained by Fish Richardson (Contact: Joseph Colaianni)
Date:	11/18 – 4/19

Case: ***Rohm Company IPR Petition***

Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.

Counsel: Retained by Fish & Richardson (Contact: Matt Colvin)

Date: 4/18 – 8/18

Case: ***Apple Inc., vs. Qualcomm Incorporated***

Jurisdiction: Case No. 3:17-CV-02402

United States District Court for the Southern District of California

Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration, ***2 depositions taken.***

Counsel: Fish & Richardson (Contact: Tony Nguyen)

Date: 1/18 – 5/19

Case: ***Macronix vs. Toshiba***

Jurisdiction: Investigation No. 337-TA-1046

United States International Trade Commission

Work Product: Testifying expert for infringement and non-invalidity on 2 patents, claim construction analysis, technical tutorial, database viewing for evidence gathering, issuing expert report and expert rebuttal report, ***deposition taken for 2 patents***, issued witness statement and rebuttal witness statement, ***testified in ITC court on both infringement and invalidity.***

Nature of Case: ITC infringement case concerning advanced NAND Flash memory.

Counsel: Retained by Fish Richardson (Contact: David Barken)

Date: 4/17 – 10/18

Case: ***Novatek Microelectronics Corp. IPR Petition***

Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.

Counsel: Retained by Kroub, Silbersher & Kolmykov PLLC (Contact: Sergey Kolmykov)

Date: 2/17 – 4/17

Case: ***Taiwan Semiconductor Manufacturing Company IPR Petition***

Work Product: Patent analysis and prior art review toward preparation of IPR petition material and draft of declaration.

Counsel: Retained by Fish & Richardson (Contact: John Goetz)

Date: 7/15 – 4/17

Case: **Triune Systems, LLC vs. Wayne Chen, et al.,**
Jurisdiction: Case No. 296-03209-2013,
District Court of Collin County, TX
Work Product: Testifying expert, database viewing, expert report, **video-taped testimony**
Nature of Case: Defending against trade secret contention.
Counsel: Retained by Gordon & Rees LLP (Contact: Robert Bragalone)
Date: 5/15 – 9/15

Case: **Macronix vs. Spansion**
Jurisdiction: Investigation No. 337-TA-922
United States International Trade Commission
Work Product: Testifying expert, claim construction analysis, technical tutorial, database viewing for evidence gathering, expert report, expert rebuttal report.
Nature of Case: ITC infringement case concerning Flash memory cell
Counsel: Retained by Fish Richardson (Contact: David Barken)
Date: 9/14 – 1/15

Case: **Macronix vs. Spansion**
Jurisdiction: Investigation No. 337-TA-909
United States International Trade Commission
Work Product: Database viewing for evidence gathering, support technical content of expert report, support technical content against rebuttal report.
Nature of Case: ITC infringement case concerning Flash memory cell
Counsel: Retained by Winston & Strawn (Contact Michael Murray)
Date: 5/14 – 1/15

Case: **Keranos, LLC v. Analog Devices et. al.**
Jurisdiction: Civil Case No. 2:10-CV-00207
United States District Court for the Eastern District of Texas
Work Product: Claim Chart Generation and Claim Construction
Nature of Case: Patent infringement case concerning split gate Flash memory cell
Counsel: Retained by Agility IP (Contact Michelle Breit)
Date: 2/09 – 2/14

Case: **Silicon Storage Technology, Inc. v. Xicor, LLC**
Jurisdiction: Civil Case No. CV 10-01515
United States District Court for the Northern District of California
Work Product: Expert report and declarations
Nature of Case: Matter concerning United States Reissue Patent RE38,370 on the process of flash memory cell.
Counsel: Retained by Shore Chan Bragalone LLP (Contact Patrick Conroy)
Date: 6/11 – 10/11

Case: ***Fast Memory Erase, LLC v. Spansion et.al.***
Jurisdiction: Civil Case No. 3:10CV-481
United States District Court for the Northern District of Texas
Work Product: Declarations
Nature of Case: Defending Motion for Attorney Fees.
Counsel: Retained by Shore Chan Bragalone LLP (Contact Patrick Conroy)
Date: 2/10 – 6/10

Case: ***Fast Memory Erase, LLC v. Spansion et.al.***
Jurisdiction: Civil Case No. 3-08cv0977M
United States District Court for the Northern District of Texas
Work Product: Testifying expert: issuing expert report, declarations, and
deposition taken.
Nature of Case: Patent infringement case concerning minimizing source current
during erase of Flash memory cell.
Counsel: Retained by Shore Chan Bragalone LLP (Contact Patrick Conroy)
Date: 2/09 – 5/10

PROFESSIONAL EXPERIENCE:

10/07 to present: **Technical Consultant/Expert Witness**

Serving as a technical consultant and advisor in the area of semiconductor device and process technologies. Expert witness in ITC cases and district court cases, with successful testifying experience in trial hearing. Depositions taken at least 12 times for both litigation cases and IPR petitions. Experienced in source code viewings of circuit schematics and GDS layouts; seasoned in reverse engineering of integrated circuits, as well as techniques for de-processing and analyzing semiconductor chips. Responsibilities including gathering and reviewing of evidence, issuing declaration on claim construction, giving technical tutorial, issuing expert report and expert rebuttal report. Technical consultant for evaluating numerous flash memory and CMOS image sensor patent portfolios for licensing and acquisition consideration. Assessed qualitative merits of each patent portfolio and identified potential infringers and potential licensees. Generated numerous claim charts against potential infringers based on published patents, technical papers and datasheets, and reverse engineering reports.

10/07 to 4/11:

Jonker, LLC

Inventor and Partner. Developed an intellectual property (IP) portfolio (7 U.S. patents granted and 3 U.S. patent applications pending) on zero-cost CMOS-logic-compatible embedded flash memory. Successfully negotiated the acquisition of this IP portfolio by a major semiconductor company.

8/04 to 10/07:

Maxim Integrated Products, San Jose, California

Senior Scientist. Responsible for developing embedded non-volatile memory process technology for battery management products. Duties included supervising the development of integrated process flows at the 0.35 μ m and 0.18 μ m CMOS technology nodes. Analyzed and solved process-related reliability issues.

5/00 to 8/04:

Xilinx, Incorporated, San Jose, California

Senior Manager. Responsible for developing non-volatile memory process technology for flash and CPLD products, as well as advanced CMOS process technology (75nm node). Duties included supervising the development of integrated non-volatile memory process flows at multiple CMOS technology nodes to meet product application requirements. Analyzed and solved process-related reliability issues.

2/00 to 5/00

Progressant Technologies, Fremont, California

Co-Founder. Facilitated IP development and sale of the company to Synopsys, Inc.

9/98 to 2/00

Programmable Silicon Solutions, Sunnyvale, California

Director of Process Engineering. Responsible for developing a fully logic compatible embedded flash memory technology for WSMC (a semiconductor foundry). Developed a new integrated process flow to implement a proprietary flash cell in a high-performance logic process, supervised testchip and product tapeout, process optimization, device optimization, and product yield enhancement. Applied interpersonal skills to align the goals of the foundry technology development group to that of PSS. Participated in the strategic planning of a technology roadmap and foundry strategy for the company.

9/97 to 9/98

AMIC Technology, Santa Clara, California

Director of Flash Technology. Responsible for developing 0.35µm ETOX-based flash memory technology. Prepared the business plan and technology roadmap. Developed a new integrated process flow, supervised testchip and product tapeout, process optimization, device optimization, and product yield enhancement.

5/96 to 9/97

Altera Corp., San Jose, California

Device Engineering Manager. Responsible for technology development, process integration, and foundry interface for 0.35µm generation of logic and EEPROM programmable logic device technology. Duties included project schedule planning, supervision of testchip and product tapeout, process optimization, and product yield enhancement.

7/95 to 5/96

Information Storage Devices, San Jose, California

Technology Development Manager/Program Manager. Led a project team of 10 in technology development, process integration, and first product introduction, with a new foundry. Duties included project schedule planning, supervision of testchip and product tapeout, process optimization, resolution of sorting issues and yield enhancement.

5/92 to 7/95

Advanced Micro Devices, Sunnyvale, California

Member of Technical Staff. Led the development of low-energy/DINOR-type flash EPROM technology. Key individual contributor in optimizing the process and design of flash cell and periphery devices in AMD's 0.5µm and 0.35µm flash EPROM technologies. Duties included process integration, device modeling, and development of triple-well technology and high-voltage transistors for negative gate erase operation.

1/89 to 4/92

Texas Instruments, Dallas, Texas

Member of Technical Staff. Research and development of 16Mb generation of flash EPROM technology, as well as a new generation of antifuse-based FPGAs. Duties included process integration, device modeling, high-voltage CMOS process integration, and investigation of a novel source-side injection mechanism for EPROM channel hot-electron programming.

AWARDS/HONORS

U.C. Berkeley Certificate of Distinction Award, 1983

AEA Faculty Development Fellowship, 1983-1985

AMD Spotlight Award, 1995

PATENTS

1. U.S. Patent 5,106,773, "Programmable Gate Array and Methods for Its Fabrication" (with K.-L. Chen and H. Tigelaar), issued April 21, 1992.
2. U.S. Patent 5,166,557, "Gate Array with Built-in Programmable Circuitry" (with K.-L. Chen), issued November 24, 1992.
3. U.S. Patent 5,202,576, "Asymmetrical Non-Volatile Memory Cell, Arrays, and Methods for Fabricating the Same" (with M. Wong), issued April 13, 1993.
4. U.S. Patent 5,219,782, "Sublithographic Antifuse and method for Manufacturing" (with K.-L. Chen), issued June 15, 1993.
5. U.S. Patent 5,250,464, "Method of Making a Low Capacitance, Low Resistance Sidewall Antifuse Structure" (with M. Wong), issued October 5, 1993.
6. U.S. Patent 5,264,384, "Method of Making a Non-Volatile Memory Cell" (with C. Kaya), issued November 23, 1993.
7. U.S. Patent 5,300,803, "Source Side Injection Non-volatile Memory Cell," issued April 5, 1994.
8. U.S. Patent 5,365,105, "Sidewall Antifuse Structure and Method for Making," issued November 15, 1994.
9. U.S. Patent 5,371,402, "Low Capacitance, Low Resistance Sidewall Antifuse Structure and Process" (with M. Wong), issued December, 1994
10. U.S. Patent 5,395,797, "Antifuse Structure and Method of Fabrication," issued March 7, 1995.
11. U.S. Patent 5,470,773, "Method of Protecting a Stacked Gate Edge in a Semiconductor Device from Self Aligned Source (SAS) Etch" (with Y. Sun and C. Chang), issued November 28, 1995.
12. U.S. Patent 5,482,880, "Non-Volatile Memory and Fabrication Method," (with C. Kaya), issued January 9 1996.
13. U.S. Patent 5,517,443, "Method and System for Protecting a Stacked Gate Edge in a Semiconductor Device from Self Aligned Source (SAS) Etch in a Semiconductor Device," issued May 14, 1996.
14. U.S. Patent 5,521,867, "Adjustable Threshold Voltage Conversion Circuit" (with Jian Chen), issued May 28, 1996.
15. U.S. Patent 5,534,455, "Method and System for Protecting a Stacked Gate Edge in a Semiconductor Device from Self Aligned Source (SAS) Etch in a Semiconductor Device," issued July 9, 1996.

16. U.S. Patent 5,541,875, "High Energy Buried Layer Implant To Provide A Low Resistance P-Well in A Flash EPROM Array" (with Jian Chen), issued May 28, 1996.
17. U.S. Patent 5,590,076, "Channel Hot-Carrier Page Write," issued December 31, 1996.
18. U.S. Patent 5,596,531, "Method for Decreasing the Discharge Time of A Flash Memory Cell," issued January 21, 1997.
19. U.S. Patent 5,612,914, "Asymmetrical Non-volatile Memory Cell, Arrays and Methods for Fabricating Same" (with Man Wong), issued May 18, 1997.
20. U.S. Patent 5,624,859, "Method for Providing Device Isolation and Off-State Leakage Current for a Semiconductor Device," issued April 29, 1997.
21. U.S. Patent 5,625,220, "Sublithographic Antifuse," (with K.L. Chen), issued April 29, 1997.
22. U.S. Patent 5,646,430, "Non-volatile Memory Cell Having Lightly-Doped Source Region" (with C. Kaya), Issued July 8, 1997.
23. U.S. Patent 5,650,964, "Method of Inhibiting Degradation of Ultra Short Channel Charge-carrying Devices during Discharge," (with Jian Chen), Issued July 22, 1997.
24. U.S. Patent 5,652,155, "Method for Making Semiconductor circuit Including Non-ESD Transistors with Reduced Degradation Due to an Impurity Implant," Issued July 27, 1997.
25. U.S. Patent 5,656,509, "Method and Test Structure for Determining Gouging in a Flash EPROM Cell During SAS Etch," Issued July 29, 1997
26. U.S. Patent 5,661,059, "Boron Penetration to Suppress Short Channel Effect in P-channel Device," Issued August 26, 1997.
27. U.S. Patent 5,674,764, "Method of Making Asymmetrical Non-volatile Memory Cell," Issued October 7, 1997.
28. U.S. Patent 5,693,972, "Method and System for Protecting a Stacked Gate Edge in a Semiconductor Device from Self-Aligned Source (SAS) Etch in a Semiconductor Device," Issued December 2, 1997.
29. U.S. Patent 5,751,631, "Flash memory cell and a new method for sensing the content of the new memory cell," Issued May 12, 1998
30. U.S. Patent 5,789,295, "Method of Eliminating or Reducing Poly1 Oxidation at Stacked Gate Edge in Flash EPROM Process," Issued August 4, 1998.
31. U.S. Patent 5,814,854, "Highly scalable FLASH EEPROM cell," Issued September 29, 1998.
32. U.S. Patent 5,814,864, "Semiconductor Circuit Including Non-ESD Transistors with Reduced Degradation Due to an Impurity Implant," Issued September 29, 1998.

33. U.S. Patent 5,912,836, "Circuit for Detecting Both Charge Gain and Charge Loss Properties in a Non-Volatile Memory Array," Issued June 15, 1999.
34. U.S. Patent 5,930,174, "Circuit and Method for Erasing Flash Memory Array," Issued July 27, 1999.
35. U.S. Patent 5,981,994, "Method and Semiconductor circuit for Maintaining Integrity of Field Threshold Voltage Requirements," Issued November 9, 1999.
36. U.S. Patent 5,995,418, "Circuit and Method for Erasing Flash Memory Array," Issued November 30, 1999.
37. U.S. Patent 6,026,017, "Compact nonvolatile memory," Issued February 15, 2000.
38. U.S. Patent 6,027,974, "Nonvolatile memory," Issued February 22, 2000.
39. U.S. Patent 6,088,263, "Non-volatile memory using substrate electrons," Issued July 11, 2000.
40. U.S. Patent 6,091,636, "Flash memory cell and a new method for sensing the content of the new memory cell," Issued July 18, 2000.
41. U.S. Patent 6,127,225, "Memory cell having implanted region formed between select and sense transistors," Issued October 3, 2000.
42. U.S. Patent 6,159,800, "Method of forming memory cell," Issued December 12, 2000.
43. U.S. Patent 6,188,604, "Flash memory cell & array with improved pre-program and erase characteristics," Issued February 13, 2001.
44. U.S. Patent 6,185,133, "Flash EPROM using junction hot hole injection for erase," Issued February 6, 2001.
45. U.S. Patent 6,252,799, "Device with embedded Flash and EERPOM memories," Issued June 26, 2001.
46. U.S. Patent 6,326,265, "Device with Embedded Flash and EPROM Memories," Issued December 4, 2001.
47. U.S. Patent 6,417,550, "High voltage MOS devices with high gated-diode breakdown voltage and punch-through voltage," Issued July 9, 2002.
48. U.S. Patent 6,479,862, "Charge trapping device and method for implementing a transistor having a negative differential resistance mode," Issued November 12, 2002.
49. U.S. Patent 6,512,274, "CMOS-process compatible, tunable NDR (negative differential resistance) device and method of operating same," Issued January 28, 2003.
50. U.S. Patent 6,596,617, "CMOS compatible process for making a tunable negative differential resistance (NDR) device," Issued July 22, 2003.

51. U.S. Patent 6,624,026, "Nonvolatile Memory," Issued September 23, 2003.
52. U.S. Patent 6,680,245, "Method for making both a negative differential resistance (NDR) device and a non-NDR device using a common MOS process," Issued January 20, 2004.
53. U.S. Patent 6,686,631, "Negative differential resistance (NDR) device and method of operating same," Issued February 3, 2004.
54. U.S. Patent 6,693,027, "Method for configuring a device to include a negative differential resistance (NDR) characteristic," Issued February 17, 2004.
55. U.S. Patent 6,700,155, "Charge trapping device and method for implementing a transistor having a configurable threshold," Issued March 2, 2004.
56. U.S. Patent 6,711,063, "EEPROM memory cell array architecture for substantially eliminating leakage," Issued March 23, 2004.
57. U.S. Patent 6,835,979, "Nonvolatile Memory," Issued December 28, 2004.
58. U.S. Patent 6,969,894, "Variable threshold semiconductor device and method of operating same," Issued November 29, 2005.
59. U.S. Patent 6,972,234, "High Voltage MOS Devices with High Gated-Diode Breakdown Voltage and Punch-through Voltage," Issued December 6, 2005.
60. U.S. Patent 6,972,465, "CMOS process compatible, tunable negative differential resistance (NDR) device and method of operating same," Issued December 6, 2005.
61. U.S. Patent 7,067,873, "Charge Trapping Device," Issued June 27, 2006.
62. U.S. Patent 7,091,077, "Method of directionally trimming polysilicon width," Issued August 15, 2006.
63. U.S. Patent 7,109,078, "CMOS compatible process for making a charge trapping device," Issued September 19, 2006.
64. U.S. Patent 7,301,194, "Shrinkable and highly coupled double poly EEPROM with Inverter," Issued November 27, 2007.
65. U.S. Patent 7,436,710, "EEPROM memory device with cell having NMOS in a P pocket as a control gate, PMOS program/erase transistor, and PMOS access transistor in a common well," Issued October 14, 2008.
66. U.S. Patent 7,535,758, "One or Multiple-Times Programming Device," Issued May 19, 2009.
67. U.S. Patent 7,782,668, "Integrated circuit embedded with Non-volatile One-Time-Programmable and Multiple-Time Programmable memory," Issued August 24, 2010.

68. U.S. Patent 7,787,295, "Integrated circuit embedded with Non-volatile Multiple-Time Programmable memory having variable coupling," Issued August 31, 2010.
69. U.S. Patent 7,787,304, "Method of making integrated circuit embedded with Non-volatile One-Time-Programmable and Multiple-Time Programmable memory," Issued August 31, 2010
70. U.S. Patent 7,787,309, "Method of operating integrated circuit embedded with Non-volatile One-Time Programmable and Multiple-Time Programmable memory," Issued August 31, 2010.
71. U.S. Patent 7,791,955, "Method of erasing a block of memory cells," Issued September 7, 2010.
72. U.S. Patent 7,835,184, "EEPROM memory cell with first-dopant-type control gate transistor, and second-dopant type program/erase and access transistors formed in common well," Issued November 16, 2010
73. U.S. Patent 7,835,186, "Method of programming a selected memory cell," Issued November 16, 2010.
74. U.S. Patent 7,852,672, "Integrated Circuit embedded with Non-Volatile programmable memory having variable coupling," Issued December 14, 2010
75. U.S. Patent 7,876,615, "Method of operating integrated circuit embedded with Non-Volatile programmable memory having variable coupling related application data," Issued January 25, 2011
76. U.S. Patent 7,920,426, "Non-volatile memory programmable through areal capacitive coupling," Issued April 5, 2011
77. U.S. Patent 7,944,750, "Multi-programmable Non-Volatile Memory Cell," Issued May 17, 2011
78. U.S. Patent 8,203,861, "Non-Volatile One-Time Programmable and Multiple-Time Programmable Device," Issued June 19, 2012
79. U.S. Patent 8,208,299, "Integrated Circuit Embedded with Non-Volatile Programmable Memory Having Variable Coupling and Separate Read/Write Paths," Issued June 26, 2012
80. U.S. Patent 8,300,470, "Two Terminal Programmable Hot Channel Electron Non-Volatile Memory," Issued October 30, 2012
81. U.S. Patent 8,305,805, "Common Drain Non-Volatile Multiple-Time Programmable Memory," Issued November 6, 2012
82. U.S. Patent 8,325,519, "Method of Operating Integrated Circuit Embedded with Non-Volatile Programmable Memory having Variable Coupling Ratio," Issued December 4, 2012.

83. U.S. Patent 8,580,622, "Method of Making Integrated Circuit Embedded with Non-Volatile Programmable Memory having Variable Coupling," Issued November 12, 2013.
84. U.S. Patent 8,582,342, "Non-Volatile One-Time-Programmable and Multiple-Time Programmable Memory Configuration Circuit," Issued November 12, 2013.
85. U.S. Patent 8,599,612, "Method of Operating Integrated Circuit Embedded with Non-Volatile Programmable Memory having Variable Coupling Related Application data," Issued December 3, 2013.
86. U.S. Patent 8,705,263, "Non-Volatile One-Time-Programmable and Multiple-Time Programmable Memory Configuration Circuit," Issued April 22, 2014.
87. U.S. Patent 8,988,103, "Capacitively Coupled Logic Gate," Issued March 24, 2015.
88. U.S. Patent 9,064,192, "Automatic Valid Vote Count Storage using Secure Embedded Non-Volatile Memory," Issued June 23, 2015.
89. U.S. Patent 9,224,739, "Method of Making Integrated Circuit Embedded with Non-Volatile Programmable Memory having Variable Coupling," Issued December 29, 2015.
90. U.S. Patent 9,305,931, "Zero Cost NVM Cell Using High Voltage Devices in Analog Process," Issued April 5, 2016.
91. U.S. Patent 9,324,438, "Method of Operating Incrementally Programmable Non-Volatile Memory," Issued April 26, 2016.
92. U.S. Patent 9,558,813, "Method of Operating Incrementally Programmable Non-Volatile Memory," Issued January 31, 2017.
93. U.S. Patent 9,570,161, "Method of Operating Incrementally Programmable Non-Volatile Memory," Issued February 14, 2017.
94. U.S. Patent 9,570,175, "Incrementally Programmable Non-Volatile Memory," Issued February 14, 2017.
95. U.S. Patent 10,061,738, "Ephemeral Peripheral Device," Issued August 28, 2018.
96. U.S. Patent 10,115,467, "One time Accessible (OTA) Non-Volatile Memory," Issued October 30, 2018.
97. U.S. Patent 10,198,889, "Automatic Valid Vote Count Storage Using Secure Embedded Non-Volatile Memory," Issued February 5, 2019
98. U.S. Patent 10,559,700, "Zero Cost ANVM Cell Using High Voltage Devices in Analog Process," Issued February 11, 2020.
99. U.S. Patent 10,629,607, "Semiconductor CMOS Non-Volatile Memory Device," Issued April 21, 2020

PUBLICATIONS

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2. D. K. Y. Liu and J. D. Plummer, "A novel trench-injector power device with low ON-resistance and high switching speed," *IEEE Electron Device Letters*, Vol. 9, No. 7, pp. 321-323, 1988.
3. D. M. Boisvert, D. K. Y. Liu and J. D. Plummer, "Circuit approaches to increasing IGBT switching speed," *IEEE Journal of Solid-State Circuits*, Vol. 23, No. 5, pp. 1276-1279, 1988.
4. D. K. Y. Liu and J. D. Plummer, "Device physics and optimization of conductivity-modulated power MOSFET's," *IEEE Transactions on Electron Devices*, Vol. 35, No. 12, pp. 2457-2458, 1988.
5. D. K. Y. Liu, K.-L. Chen, H. Tigelaar, J. Paterson and S. O. Chen, "Scaled dielectric antifuse structure for field-programmable gate array applications," *IEEE Electron Device Letters*, Vol. 12, No. 4, pp. 151-153, 1991.
6. M. Wong, D. K. Y. Liu, M. M. Moslehi and D. W. Reed, "Preoxidation treatment using HCl/HF vapor," *IEEE Electron Device Letters*, Vol. 12, No. 8, pp. 425-426, 1991.
7. D. K. Y. Liu, C. Kaya, M. Wong, J. Paterson and P. Shah, "Optimization of a source-side injection FAMOS cell for flash EPROM applications," *International Electron Devices Meeting Technical Digest*, pp. 315-318, 1991.
8. K.-L. Chen, D. K. Y. Liu, G. Misium, W. M. Gosney, S.-J. Wang, J. Camp and H. Tigelaar, "A sublithographic antifuse structure for field-programmable gate array applications," *IEEE Electron Device Letters*, Vol. 13, No. 1, pp. 53-55, 1992.
9. K. T. San, C. Kaya, D. K. Y. Liu, T. P. Ma, and P. Shah, "A new technique for determining capacitive coupling coefficients in flash EPROMs," *IEEE Electron Device Letters*, Vol. 13, No. 6, pp. 328-331, 1992.
10. C. Kaya, D. K. Y. Liu, J. Paterson and P. Shah, "Buried source-side injection (BSSI) for flash EPROM programming," *IEEE Electron Device Letters*, Vol. 13, No. 9, pp. 465-467, 1992.
11. M. Wong, D. K. Y. Liu and S. S.-W. Huang, "Analysis of the subthreshold slope and the linear transconductance techniques for the extraction of the capacitance coupling coefficients of floating-gate devices," *IEEE Electron Device Letters*, Vol. 13, No. 11, pp. 566-568, 1992.
12. D. K. Y. Liu and J. D. Plummer, "Design and analysis of a new conductivity-modulated power MOSFET," *IEEE Transactions on Electron Devices*, Vol. 40, No. 2, pp. 428-438, 1993.
13. J. Z. Peng, S. Haddad, H. Fang, C. Chang, S. Longcor, B. Ho, Y. Sun, D. Liu, Y. Tang, J. Hsu, S. Luan, J. Lien, "Flash EPROM endurance simulation using physics-based models," In *International Electron Devices Meeting Technical Digest*, pp. 295-298, 1994.

14. J. Chen, J. Hsu, S. Luan, Y. Tang, D. Liu, S. Haddad, C. Chang, S. Longcor, J. Lien, "Short Channel enhanced degradation during discharge of Flash EEPROM memory Cell," *International Electron Devices Meeting Technical Digest*, pp. 331-334, 1995.
15. V. H. Chan and D. Liu, "An Enhanced Erase Mechanism During Channel Fowler-Nordheim Tunneling in Flash EPROM Memory Devices," *IEEE Electron Devices Letters*, Vol. 20, No. 3, pp. 140-142, 1999.

EXHIBIT B

Jack C. Lee
Professor Emeritus
Electrical and Computer Engineering Department
The University of Texas at Austin

Current Status

Professor Emeritus
Department of Electrical and Computer Engineering
The University of Texas at Austin

Research / teaching experience and expertise:

- Integrated circuits manufacturing
- Photovoltaic/solar cell technologies
- Alternative channel devices
- Dielectric and passivation processes
- Display technologies
- Semiconductor memory devices

Education

9/84 – 8/88 Ph.D., Electrical Engineering, University of California at Berkeley
9/80 – 12/81 M.S., Electrical Engineering, University of California at Los Angeles
9/76 – 6/80 B.S., Electrical Engineering (with highest honors),
University of California at Los Angeles

Professional Career

9/23 – present Professor Emeritus
Department of Electrical and Computer Engineering
The University of Texas at Austin
9/00 – 8/23 Cullen Trust for Higher Education Endowed Professorship in Engineering #4
Department of Electrical and Computer Engineering
The University of Texas at Austin
9/96 – 8/23 Professor, Department of Electrical and Computer Engineering
The University of Texas at Austin
9/92 – 8/96 Associate Professor, Department of Electrical and Computer Engineering
The University of Texas at Austin
9/88 – 8/92 Assistant Professor, Department of Electrical and Computer Engineering
The University of Texas at Austin
1/88 – 6/88 Lecturer, Electrical Engineering and Computer Science Department
University of California at Berkeley
6/79 – 8/84 Member of Technical Staff, High-Speed Bipolar Device Program
TRW Microelectronics Center, Redondo Beach, CA

Honors and Awards

IEEE Electron Devices Society Distinguished Lecturer, 2004 - 2016.
Fellow, The Institute of Electrical and Electronic Engineers (IEEE), 2002 “For
contributions to the understanding and development of ultra-thin dielectrics and their
application to silicon devices”
Gordon Lepley IV Endowed Memorial Teaching Award, ECE Department, The
University of Texas at Austin, 2004

Cullen Trust For Higher Education Endowed Professorship in Engineering, 2000-present
 Dean's Fellow, College of Engineering, The University of Texas at Austin, 1999, 2003
 Lockheed Fort-Worth Division Award for Excellence in Engineering Teaching,
 College of Engineering, The University of Texas at Austin, 1996
 Award of Excellence, Halliburton Foundation, 1993
 Departmental Teaching Award, College of Engineering, The University of Texas at
 Austin, 1993
 SRC Inventor Recognition Award, Semiconductor Research Corporation, 1991
 Hughes Aircraft Company Endowed Faculty Fellowship in Engineering,
 The University of Texas at Austin, 1991- 2000
 Outstanding Engineering Teaching by an Assistant Professor, College of Engineering,
 The University of Texas at Austin, 1991
 Best Paper Award, SEMATECH Centers of Excellence Coordination Meeting, 1990.
 Dow Outstanding Young Faculty Award, American Society for Engineering Education,
 1990
 Engineering Research Initiation Award, Engineering Foundation of the United
 Engineering Trustees, 1989
 Best Paper Award, IEEE International Reliability Physics Symposium, 1988

Litigation Experience as Testifying Experts

The following is a list of cases in which I have provided expert testimony, including through a declaration, report or testimony at deposition or trial since 2016

- HD Silicon Solutions LLC v. NXP USA
 July 2026 –
 (Representing defendant, NXP)
- Marlin Semiconductor Limited v. TSMC
 December 2025 –
 (Representing defendant, TSMC)
- Longitude Flash Memory Solutions and IPValue Management v. SanDisk
 April 2025 – February 2026
 (Representing defendant, SanDisk)
- Marlin Semiconductor Limited v. TSMC
 April 2025 –
 (Representing defendant, TSMC)
- Advanced Integrated Circuit Process LLC v. Taiwan Semiconductor Mfg. Co. Ltd
 October 2024 –
 (Representing defendant, TSMC)
- Yangtze Memory Technologies v. Micron
 January 2024 –
 (Representing defendant, Micron)

- BeSang v. Micron
March 2023 –
(Representing defendant, Micron)
- The Trustees of Purdue University v. Wolfspeed, Inc.
February 2022 – March 2025
(Representing defendant, Wolfspeed)
- GlobalFoundries - IPR
March 2021 –
(Representing petitioner and filed an IPR declaration)
- HD Silicon Solutions LLC v. Microchip Technology Inc.
January 2021 – February 2025
(Representing defendant, Microchip Technology Inc.)
- Longitude Licensing Limited vs Western Digital
March 2023 – January 2024
(Representing defendant, Western Digital)
- Impinj, Inc. v. NXP USA, Inc.
September 2019 – July 2023
(Representing defendant, NXP USA, Inc.)
- Monterey Research, LLC v. Qualcomm Inc.
February 2020 – May 2022
(Representing defendant, Qualcomm)
- Bell Semiconductor, LLC v. Texas Instruments Incorporated
July 2020 – March 2021
(Representing defendant, Texas Instruments)
- Greenthread LLC v. Samsung
February 2020 – July 2020
(Representing plaintiff, Greenthread LLC)
- MLC Intellectual Property LLC vs. Micron Technology, Inc.,
June 2016 – January 2021
(Representing plaintiff, MLC Intellectual Property LLC)
- Hanwha v. LONGi Solar Technology
April 2019 – June 2020
(Representing defendant LONGi Solar Technology)
- Vista Peak Ventures, LLC v. Innolux Corp.

January 2019 – March 2019
(Representing IPR Petitioner, Innolux Corp.)

- Home Semiconductor Corp. v. Samsung
August 2018 – June 2020
(Representing plaintiff, Home Semiconductor Corp.)
- Mariner IC Inc. v. Asustek Computer Inc.
May 2018 – August 2019
(Representing plaintiff, Mariner IC Inc.)
- Godo Kaisha IP Bridge 1 v. OmniVision Technologies, Inc.
January 2018 – December 2018
(Representing defendant, OmniVision Technologies, Inc.)
- Macronix v. Toshiba
May 2017 – February 2018
(Representing defendant, Toshiba)
- University of Illinois v. Micron Technology
September 2009 – November 2017
(Representing plaintiff, Univ. of Illinois)
- Silicon Genesis v. Soitec
October 2016 – March 2017
(Representing plaintiff, Silicon Genesis)

Non-Litigation Consulting

- Applied Materials, Inc.
Teaching short courses on semiconductor physics, devices, design/layout, and fabrication technologies (2-3 courses per year).

Jack C. Lee is a Professor Emeritus of the Electrical and Computer Engineering Department. He held the Cullen Trust for Higher Education Endowed Professorship in Engineering # 4 at The University of Texas at Austin from 2000-2023. He received the B.S. and M.S. degrees in electrical engineering from University of California, Los Angeles, in 1980 and 1981, respectively; and the Ph.D. degree in electrical engineering from University of California, Berkeley, in 1988. From 1979 to 1984, he was a Member of Technical Staff at the TRW Microelectronics Center, CA, in the High-Speed Bipolar Device Program. He worked on bipolar device and circuit design, fabrication and testing. In 1988, he joined the faculty of The University of Texas at Austin. His current research

interests include semiconductor devices including 3D structures such as FinFETs, fabrication processes, characterization and modeling, dielectric process, characterization and reliability, high-K gate dielectrics and electrode, semiconductor memory applications, alternative channel materials, display technologies and photovoltaic/solar cell technologies. Dr. Lee has over 40 years of experience in semiconductor technology and dielectric processing. He has published over 600 journal publications and conference proceedings and 9 issued US patents; and co-authored 8 book and book chapters. He has also been recognized with many teaching and research awards. Dr. Lee is a Fellow of IEEE and had been a Distinguished Lecturer for the IEEE Electron Devices Society.

Publications

Jack C. Lee

Refereed Journal Publications

1. J. Lee, K. Mayaram and C. Hu, "A Theoretical Study of Gate/Drain Offset in LDD MOSFET's," IEEE Electron Device Letters, vol. EDL-7, no. 3, p. 152 - 154, March 1986.
2. J. Lee, I-C Chen and C. Hu, "Comparison Between CVD and Thermal Oxide Dielectric Integrity," IEEE Electron Device Letters, vol. EDL-7, no. 9, p. 506 - 509, September 1986.
3. K. Mayaram, J. Lee and C. Hu, "A Model for the Electric Field in Lightly Doped Drain Structures," IEEE Transactions on Electron Devices, vol. ED-34, no.7, p. 1509 - 1518, July 1987.
4. J. Lee and C. Hu, "Polarity Asymmetry of Oxides Grown on Polycrystalline Silicon," IEEE Transactions on Electron Devices, vol. ED-35, no. 7, p. 1063 - 1070, July 1988.
5. J. Lee, C. Hegarty and C. Hu, "Electrical Characteristics of MOSFET's Using Low-Pressure Chemical Vapor Deposited Oxide," IEEE Electron Device Letters, vol. EDL-9, no. 7, p. 324 - 327, July 1988.
6. J. Lee, I-C Chen and C. Hu, "Modeling and Characterization of Gate Oxide Reliability," Special Issue of IEEE Transactions on Electron Devices on Reliability, Vol. ED-35, no. 12, p. 2268 - 2278, December 1988.
7. H. Hwang, W. Ting, D. L. Kwong, J. Lee, L. Buhrow and R. Bowling, "Electrical Characteristics of Reoxidized-nitrided CVD Oxide," Applied Physics Letters, 55(8), p. 755 - 756, 21 August, 1989.
8. R. Moazzami, J. Lee and C. Hu, "Temperature Acceleration of Time-Dependent Dielectric Breakdown," Special issue of IEEE Transactions on Electron Devices on Vacuum Microelectronic Devices, vol. ED-36, no. 11, p. 2462 - 2465, November 1989.
9. H. Hwang, W. Ting, D.L. Kwong, J. Lee, L. Buhrow and R.A. Bowling, "Effects of Dynamic Stressing on Nitrided and Reoxidized-Nitrided Chemical Vapor Deposited Gate Oxides," IEEE Electron Device Letters, vol. EDL-10, no. 12, p. 568 - 570, December 1989.
10. J. Lin, S. Banerjee, J. Lee, and C. Teng, "Soft Breakdown in Titanium-Silicided Shallow Source/Drain Junction," IEEE Electron Device Letters, vol. EDL-11, no. 5, p. 191 - 193, May 1990.
11. K. Park, S. Batra, J. Lin, S. Yoganathan, J. Lee, S. Banerjee, S. Sun, J. Yeargain, and G. Lux, "Anomalous Capacitance-Voltage Behavior Due to Dopant Segregation and

Carrier Trapping in Arsenic-Implanted Polysilicon and Polycide Gates," Applied Physics Letters, vol. 56, no. 23, p.2325 - 2327, June 4, 1990.

12. J. Lin, S. Banerjee, J. Lee, and C. Teng, "Anomalous Current-Voltage Behavior in Titanium-Silicided Source/Drain Junctions," Journal of Applied Physics, vol. 68, no. 3, p. 1082 - 1087, August 1, 1990.
13. H. Hwang, W. Ting, B. Maiti, D.L. Kwong and J. Lee, "Electrical Characteristics of Ultrathin Gate Dielectrics Prepared by Rapid Thermal Oxidation of Si in N₂O," Applied Physics Letters, vol. 57, no. 10, p. 1010 - 1011, Sept. 3, 1990.
14. S. Batra, K. Park, J. Lin, S. Yoganathan, J. Lee, S. Banerjee, S. Sun, J. Yeargain and G. Lux "Effects of Dopant Redistribution, Segregation and Carrier Trapping in As-Implanted MOS Gates," IEEE Transactions on Electron Devices, vol. 37, no. 11, p. 2322 - 2330, October, 1990.
15. S. Bhattacharya, S. Banerjee, J. Lee, A. Tasch and A. Chatterjee, "The Impact of Trench Isolation on Latch-up Immunity in Bulk, Non-epitaxial CMOS," IEEE Electron Device Letters, vol. EDL-12, no. 2, p. 77 - 79, February, 1991.
16. W. Ting, H. Hwang, J. Lee and D. L. Kwong, "Composition and Growth Kinetics of Ultrathin SiO₂ Films Formed by Oxidizing Si Substrate in N₂O," Applied Physics Letters, vol. 57, p. 2808 - 2810, 1990.
17. W. Ting, P.C. Li, G. Q. Lo, J. Lee and D.L. Kwong, "Metal-Oxide Semiconductor Characteristics of Rapid Thermal Processed Chemical Vapor Deposited SiO₂ Gate Dielectrics," Solid State Electronics, vol. 34, no. 4, p 385 - 388, 1991.
18. W. Ting, H. Hwang, J. Lee and D.L. Kwong, "Growth Kinetics of Ultrathin SiO₂ Films Prepared by Rapid Thermal Oxidation of Si Substrates in N₂O," Journal of Applied Physics, vol. 70, no. 2, p. 1072 - 1074, July 15, 1991.
19. J. Lin, K. Park, S. Batra, S. Banerjee, J. Lee and G. Lux, "Enhancement of Boron Diffusion Through Gate Oxides in MOS Devices During Rapid Thermal Silicidation," Applied Physics Letters, vol. 58, no. 19, p. 2123 - 2125, May 1991.
20. H. Hwang, W. Ting, D. L. Kwong and J. Lee, "A Physical Model for Boron Penetration Through Oxynitride Gate Dielectric Prepared by Rapid Thermal Processing in N₂O," Applied Physics Letters, vol. 59, no. 13, p. 1581 - 1582, September 23, 1991.
21. H. Hwang, W. Ting, D. L. Kwong and J. Lee, "Improved Reliability Characteristics of Submicron nMOSFET's with Oxynitride Gate Dielectrics Prepared by Rapid Thermal Oxidation in N₂O," IEEE Electron Device Letters, vol. 12, no.9, p.495-497, September 1991.

22. J. Carrano, C. Sudhama, V. Chikarmane, J. Lee, A. Tasch, W. Shepherd and N. Abt, "Electrical and Reliability Properties of PZT Thin Films for ULSI DRAM Applications," IEEE Transactions on Ultrasonics, Ferroelectrics and Frequency Control, vol. 38, no. 6, p. 690 - 703, November 1991.
23. V. Chikarmane, C. Sudhama, J. Kim, J. Lee, A. Tasch and S. Novak, "Comparative Study of The Perovskite Phase Microstructure Evolution and Electrical Properties of PZT Thin Film Capacitors Annealed in Oxygen and Nitrogen Ambients," Applied Physics Letters, vol. 59, no. 22, p. 2850 - 2852, Nov. 25, 1991.
24. M. Hao and J. Lee, "Electrical Characteristics of Oxynitrides Grown on Textured Single-Crystal Silicon," Applied Physics Letters, vol. 60, no. 4, p. 445 - 447, January 27, 1992 .
25. S. Bhattacharya, S. Banerjee, J. Lee, A. Tasch and A. Chatterjee, "Parametric Study of Latchup-Immunity of Deep Trench-Isolated, Bulk, Non-epitaxial CMOS," IEEE Trans. Elec. Dev., vol. 39, no. 4, p. 921 - 931, April 1992.
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309. (Invited) Ying-Chen Chen, Xiaohan Wu, MeiqiGuo, Fei Zhou, Yao-Feng Chang, Burt Fowler, Chih-Hung Pan, Ting-Chang Chang, Jack C Lee, "Memcomputing (Memristor+ Computing) in Intrinsic SiO_x-Based Resistive Switching Memory", 230th Meeting of The Electrochemical Society (2016).
310. Ying-Chen Chen, Xiaohan Wu, Meiqi Guo, Fei Zhou, Yao-Feng Chang**, Burt Fowler, Chih-Hung Pan, Ting-Chang Chang, and Jack C. Lee, "Memcomputing (Memristor + Computing) in Intrinsic SiO_x-Based Resistive Switching Memory" PRiME 2016 (Invited Talk).
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313. Y.-C. Chen, YF Chang, CY Lin, X Wu, G Xu, B Fowler, TC Chang, JC Lee, "Built-in Nonlinear Characteristics of Low Power Operating One-Resistor Selector-Less RRAM By Stacking Engineering", 232nd Meeting of The Electrochemical Society (2017).
314. Jia Chen, Sungjun Kim, Ying-Chen Chen, Min-Hwi Kim, Yi Li, Xiang-Shui Miao, Yao-Feng Chang, Byung-Gook Park, Jack C. Lee, "Synaptic Properties Considering Temperature Effect in HfO_x-Based Memristor – Demonstration of Homo-thermal Synaptic Behaviors", International Symposium on VLSI Technology, Systems and Applications (VLSI-TSA) (2018).
315. Ying-Chen Chen, Xiaohan Wu, Yao-Feng Chang, Jack C Lee, "Nonlinearity Enhancement by Positive Pulse Stress in Multilevel Cell Selectorless RRAM Applications", 76th Device Research Conference (DRC). 2018.

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318. Ying-Chen Chen, Chih-Yang Lin, Yao-Feng Chang and Jack C. Lee, "Energy Efficient Nonlinear Selectorless RRAM by Current Sweep Operation for Crossbar Array Applications," IEEE Semiconductor Interface Specialists Conference (SISC), San Diego CA, December 5-8, 2018.
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322. Xiaohan Wu, Ruijing Ge, Po-An Chen, Meng-Hsueh Chiang, Deji Akinwande, Jack C. Lee, "Non-Volatile Resistance Switching Phenomenon in Monolayer h-BN", 236th Electrochemical Society ECS Meeting, 2019. (Invited talk)
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328. Xiaohan Wu, Ruijing Ge, Myungsoo Kim, Deji Akinwande and Jack C. Lee, "Atomritors: Non-volatile Resistance Switching in 2D monolayers", Proceedings of the 2020 Pan Pac Microelectronics Symposium.
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333. Xiaohan Wu, Ruijing Ge, Yuqian Gu, Saban M. Hus, Jack C. Lee and Deji Akinwande "A Library of Atomritors: Non-Volatile Resistive Switching Phenomenon in 2D Atomic Sheets", Device Research Conference 2021. (Invited talk)
334. Xiaohan Wu, Ruijing Ge, Yuqian Gu, Emmanuel Okogbue, Jianping Shi, Abhay Shivayogimath, Peter Bøggild, Timothy J. Booth, Yanfeng Zhang, Yeonwoong Jung, Jack C. Lee and Deji Akinwande "Universal Non-Volatile Resistive Switching Behavior in 2D Metal Dichalcogenides Featuring Unique Conductive-Point Random Access Memory Effect", Electron Devices Technology and Manufacturing Conference (EDTM) 2021.

335. Xiaohan Wu, Ruijing Ge, Myungsoo Kim, Deji Akinwande and Jack C. Lee “Atomristors: Non-volatile Resistance Switching Behavior in 2D Monolayers”, *TechConnect World Innovation Conference 2021*.
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338. Y. Huang, Y. Gu, X. Wu, R. Ge, Y.-F. Chang, X. Wang, J. Zhang, D. Akinwande and J. C. Lee, “ReSe₂-based RRAM and Circuit-level Model for Neuromorphic Computing”, *Frontiers in Nanotechnology*, 89, 2021.
339. Y. Huang, X. Wu, Y. Gu, R. Ge, J. Zhang, Y.-F. Chang, D. Akinwande and J. C. Lee. “2D ReSe₂ RRAM and Analog Model for Neuromorphic Computing”. *52nd IEEE Semiconductor Interface Specialists Conference (SISC)*, 2021.
340. Y. Huang, Y. Gu, Y.-F. Chang, D. Akinwande, J. C. Lee. “Understand the Resistive Switching Mechanism of 2D RRAM by Monte Carlo Modeling”. *53rd IEEE Semiconductor Interface Specialists Conference (SISC)*, 2022.

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2. M. Gardner, M. Gilmer, J. Fulford and J. Lee, "Semiconductor Device Having a Tri-Layer Gate Insulating Dielectric," (issued May 2000, US 6,057,584)
3. M. Gardner, J. Fulford and J. Lee, "Semiconductor Having a PMOS Device Having Source/Drain Region Formed using a Heavy Atom P-type Implant and Method of Manufacturing Thereof," (issued January 11, 2000, US 6,013,546).
4. M. Gardner, J. Fulford and J. Lee, “Semiconductor Device with Asymmetric PMOS Source/Drain Implant and Method of Manufacture Thereof” (issued Nov. 14, 2000, US 6,146,934)
5. B. Doyle and J. Lee, “Method for Forming a High Dielectric Constant Insulator in the Fabrication of an Integrated Circuit” (issued Oct. 2001, US6,306,742)
6. Jack Lee and Han Zhao, “Tunnel Field Effect Transistor (TFET) with Lateral Oxidation,” (issued Mar. 26, 2016, US 9,293,591).
7. Jack Lee and Han Zhao, “Tunnel Field Effect Transistor (TFET) with Lateral Oxidation,” (issued Dec. 26, 2017, US 9,853,135).
8. Jack C. Lee and Fei Xue, “Three-Dimensional Gate-Wrap-Around Field Effect Transistor” (issued June 2, 2015, US 9,048,330)

9. Jack C. Lee and Fei Xue, "Vertical III-V Nanowire Field-Effect Transistor Using Nanosphere Lithography," (issued Dec. 8, 2015, US 9,209,271).
10. Deji Akinwande, Xiaohan Wu, Ruijing Ge and Jack C. Lee, Non-volatile Resistance Switching in Monolayer Atomic Sheets. *Serial No. 62/548,295, Aug. 2017*

Book / Book Chapters

1. Xiaohan Wu, Ruijing Ge, Deji Akinwande, Jack C. Lee "Memristors based on 2D monolayer materials", Memristors. Published by Intech 2021.
2. Ruijing Ge, Xiaohan Wu, Myungsoo Kim, Jack C. Lee and Deji Akinwande, "2D Materials based Non-volatile Resistive Memories and RF Switches", *Emerging 2D Materials and Devices for the Internet of Things: Information, Sensing and Energy Applications*, 1st Edition. ISBN: 978-012-81-8386-1, Published by Elsevier, June 2020.
3. Cheng Chih Hsieh, Yao-Feng Chang, Ying-Chen Chen, Chih-Yang Lin, Meiqi Guo, Fei Zhou, Burt Fowler, Ting-Chang Chang, Sanjay Banerjee, Jack C. Lee, "Memristor and Memristive Neural Networks: Review of Recently Progress on Memcomputing (Memristor + Computing) Applications in Intrinsic SiO_x-Based Resistive Switching Memory", ISBN: 978-953-51-5481-5, InTech (2018).
4. Yao-Feng Chang, Burt Fowler, Ying-Chen Chen, Fei Zhou, Kuan-Chang Chang, Tsung-Ming Tsai, Ting-Chang Chang, Simon M. Sze, and Jack C. Lee, "A Synaptic Device Built in One Diode-One Resistor (1D-1R) Architecture with Intrinsic SiO_x-based Resistive Switching Memory", *Nano Devices and Sensors*, Published by De Gruyter publisher, March (2016).
5. Yao-Feng Chang, Burt Fowler, Ying-Chen Chen, Fei Zhou, Kuan-Chang Chang, Tsung-Ming Tsai, Ting-Chang Chang, Simon M. Sze, and Jack C. Lee, "A Synaptic Device Built in One Diode-One Resistor (1D-1R) Architecture with Intrinsic SiO_x-based Resistive Switching Memory", *Nano Devices and Sensors*, p. 91-112, ISBN: 978-1-5015-0153-1, Published by De Gruyter, (March 2016).
6. Yao-Feng Chang, Burt Fowler, Ying-Chen Chen, Fei Zhou, Xiaohan Wu, Yen-Ting Chen, Yanzhen Wang, Fei Xue, and Jack C. Lee, "Resistive Switching Characteristics and Mechanisms in Silicon Oxide Memory Devices," *Nano Devices and Sensors*, p. 73-90, ISBN: 978-1-5015-0153-1, Published by De Gruyter publisher, (March 2016).
7. Y. H. Kim and J. Lee, "Hf-Based High-K Dielectrics, Process Development, Performance Characterization and Reliability," ISBN: 9781598290042, Morgan and Claypool Publishers (Jan. 2006).
8. K. Onishi and J. Lee, "High-*k* Transistor Characteristics" in the book entitled, "High *k* Gate Dielectrics" (Ed. Michel Houssa), ISBN:0-7503-0906-7, published by IOP Publishing Ltd (2004).

EXHIBIT C

CURRICULUM VITA

MASSOUD PEDRAM

Personal Information

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University of Southern California
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Wiki: https://en.wikipedia.org/wiki/Massoud_Pedram

Impact on the Field

- Advancing the theory and practice of energy-efficient computing and introducing methodologies, hardware/software solutions, and electronic design automation algorithms and tools for increasing computation speed and reducing power dissipation in CMOS circuits and systems, custom computing architectures, power electronics including power conversion and regulation, semiconductor and superconductor manufacturing, modeling and analysis of semiconductor devices including bulk and FinFet devices, superconductor electronics and superconductive computing, and quantum computing,

Research Interests

- Integrated circuit design, Computer-aided design methodologies and tools for CMOS circuits and systems, high-performance and energy-efficient computing, power/thermal management, power conversion and regulation, domain-specific accelerator design, semiconductor device design and manufacturing, electrical energy storage systems, embedded systems, cloud computing and data center resource management, homomorphic computing, post-CMOS and beyond-CMOS devices and circuits including 3D integrated circuits, superconductor electronics, and quantum computing hardware.

Education

- University of California at Berkeley, Ph.D. in Electrical Engineering and Computer Sciences, 1991
- University of California at Berkeley, M.S. in Electrical Engineering and Computer Sciences, 1989
- California Institute of Technology, B.S. in Electrical Engineering, 1986

Employment History

- Charles Lee Powell Chair in Electrical Engineering and Computer Science in the Viterbi School of Engineering, University of Southern California, 2017-present
- Stephen and Etta Varra Professor in the Viterbi School of Engineering, University of Southern California, 2013-2017

- Professor, Department of Electrical Engineering, University of Southern California, 2000-2012
- Associate Professor, Department of Electrical Engineering, University of Southern California, 1996-2000
- Assistant Professor, Department of Electrical Engineering, University of Southern California, 1991-1996
- Part-time Research Position, Xerox Palo Alto Research Center, 1987-1989
- Graduate Student Researcher, Department of Electrical Engineering and Computer Sciences, University of California at Berkeley, 1986-1991

Honors and Awards

- Elected a *Fellow of the National Academy of Inventors (Class of 2025)*, an honor that recognizes academic inventors whose work has generated significant societal and economic impact.
- Recipient of the 2023 *IEEE Trans. on Applied Superconductivity Van Duzer Prize Award*
- "For contributions to low power design of VLSI circuits and contributions to energy efficient computing," *ACM Fellow*, 2023
- "For leadership in low-power design of VLSI circuits and contributions to energy-efficient computing resulting in sustainable computing infrastructure," Fellow of the *American Association for the Advancement of Science (AAAS)*, 2023
- Best Paper Award of the *Int'l Symp. on Quality Electronic Design*, Mar. 2020
- Best Paper Award of the *Asia and South Pacific Design Automation Conference*, Jan. 2019
- *USC Viterbi School of Engineering Senior Research Award*, 2017
- "For contributions to modeling and design of low power VLSI circuits and systems, and energy efficient computing," *IEEE Circuits and Systems Society Charles A. Desoer Technical Achievement Award*, 2015
- Recognized as the *Second Most Prolific and Second Most Cited Author Awards* from the 20th Anniversary *Int'l Symp. on Low Power Electronics and Design*, Rome, Italy, Jul. 2015
- *Frequent Author Award* (Top Three Author Award) at the 20th Anniversary *Asia and South Pacific Design Automation Conference*, Chiba/Tokyo, Japan, Jan. 2015
- *ACM SIGDA Distinguished Service Award*, 2014
- Best Paper Award of the *ACM/IEEE Int'l Symp. on Low Power Design and Electronics*, Aug. 2014
- Best Paper Award of the *IEEE Computer Society Annual Symp. on VLSI*, Jul. 2014
- Recognized as one of the four DAC Prolific Authors (with 50+ papers) and the DAC Bronze Cited Author at the 50th anniversary of the Design Automation Conf., Austin, TX, Jun. 2013
- 2013 DAC Volunteer Service Award sponsored by IEEE CEDA
- Best Paper Awards of the 1989, 2000, and 2012 IEEE Int'l Conference on Computer Design, VLSI in Computers and Processors
- ACM Distinguished Scientist, 2008
- ACM SIGDA Service Award, 2004
- "For contributions to the theory and practice of low-power design and CAD," IEEE Fellow, 2001
- ACM SIGDA Distinguished Service Award, 2000
- Presidential Early Career Award for Scientists and Engineers, 1996
- NSF's Young Investigator Award, 1994
- NSF's Research Initiation Award, 1992
- 2007 IEEE Trans. on Circuits and Systems Guillemain-Cauer Best Paper Award
- Honorable Mention Award at the 2008 Int'l Symp. on Low Power Electronics and Design
- Best Paper Awards of the 33rd ACM/IEEE Design Automation Conference, Jun. 1996 and the 42nd ACM/IEEE Design Automation Conference, Jun. 2005
- Distinguished Paper Citation of the IEEE Int'l Conference on Computer Aided Design, 1990

- 1996 IEEE Circuits and Systems Society VLSI Systems Trans. Best Paper Award

University Service

- Co-organized and co-chaired the USC Symposium on the Future of Computing: A 25-Year Vision, March 13-14, 2025--see <https://sac.usc.edu/usc-symposium-on-the-future-of-computing/>.
- Member of the Computer Engineering Faculty Recruitment Committee, 2010-2011, 2013/2014, 2014/2015, 2016-2017, 2019-2025.
- Member of the VSoE Research Committee at USC, Sept. 2013 - 2017
- Member of the VSoE Transformative Faculty Hire Committee, Sept. 2013 - 2016
- Member of the Ming Hsieh Institute's Advisory Council, Jul. 2013 – Dec. 2015
- Chair of a faculty Promotion Committee in the EE department, 2014/2015, 2016-2017
- Member of the EE Dept. Faculty Recruitment Committee, 2014/2015
- Served on the Best Dissertation Selection Committee in the EE Dept., 2013
- Internal reviewer for an NSF Career Program proposal, 2013
- Chaired committee for evaluating Janet Roveda's Recruitment as an associate professor, 2012
- Division Director of USC Computer Engineering Division, Dec. 2005-Jun. 2010, and Acting Division Director at the end of 2012 and early 2013
- Served as the Chair of the 2005 Distinguished Lecture Series of the EE Dept.
- Served as the Associate Director of the CENG division of the EE Department at USC from 1998 to 2001
- Served on various departmental committees including the Faculty Merit Review Committee, Faculty Recruitment Committee, and the Promotion Committees in the EE department
- Chaired the Outstanding Research Paper Award Committee of the EE-Systems Department at USC in 1999
- Served on the Graduate Student Admissions Committee for entering M.S. and Ph.D. students in the CENG division of the EE-Systems Department at USC

Professional Service

- Served as technical area lead and panel moderator for the NSF Workshop on Micro/Nano Circuits and Systems Design in Dec 2020
- Participated in and contributed to the Computing Community Consortium (CCC) "Next Steps in Quantum Computing: Computer Science's Role," May 22-23, 2018
- Associate Editor of *Research, A Science Partner Journal*, 2018-present
- Served on the IEEE CASS Fellow (Selection) Committee, 2018
- Executive Committee Chair of the *Int'l Symp. on Low Power Electronics and Design*, July 2012 - present
- Editor-in-Chief of the *ACM Trans. on Design Automation of Electronic Systems*, Jun. 2008–May 2014
- Inaugural Editor-in-Chief of the *IEEE Journal on Emerging and Selected Topics in Circuits and Systems*, Jan. 2010–Dec. 2013
- Served as a member of the 2011 Selection Committee for the ACM SIGDA Outstanding New Faculty Award
- Served as the Chair of the Selection Committee for the IEEE CEDA and ACM SIGDA 2010 A. *Richard Newton Technical Impact Award in Electronic Design Automation*
- General Co-chair of the 2010 *IEEE Int'l Symp. on Circuits and Systems*, Paris, France
- Associate Editor
 - Foundations and Trends in Electronic Systems and Circuits Journal, Now Publishers, Inc. of Delft, 2006-2014
 - Journal of Low Power Electronics, 2005-2014

- IEEE Trans. on Computer Aided Design, 1997-2006
- IEEE Trans. on Circuits and Systems I, 2002-2003
- ACM Trans. on Design Automation of Electronic Systems, 1998-2004
- Distinguished Lecturer/Speaker
 - Association for Computing Machinery (ACM), 2007-2015
 - IEEE Solid State Circuits Society, 2001-2007
- VP of Publications of the IEEE Circuits and Systems Society, 2005-2006
- Chair of the Distinguished Lecturer Program, IEEE Circuits and Systems Society, 2003-2004
- Member of the Board of Governors, IEEE Circuits and Systems Society, 2000-2003
- Member of the ACM-SIGDA Advisory Board, managing the ACM Outstanding Ph.D. Dissertation Award in Electronic Design Automation, the DAC Young Student Support Program, the Multimedia Monograph Series, and the SIGDA Technical Committees, 1996-2009
- Chair of the Best Paper Award Committee for the IEEE Trans. on Computer Aided Design, 1999
- Member of the Research or Technical Advisory Board
 - Atrenta Inc., 2003-2015
 - Envis Inc., 2006-2009
 - PwrLite Inc., 2007-2009
 - Fulcrum Microsystems, 2001-2004
 - Aplus Design Technologies Inc., 2000-2003
 - Magma Design Automation Inc., 1997-2002
 - EPIC Design Technology, 1994-1997
- General Chair
 - *Int'l Symp. on Circuits and Systems*, June 2010
 - *Int'l Symp. on Quality of Electronic Design*, San Jose, CA, Apr. 2007
 - *Int'l Symp. on Physical Design*, Monterey, CA, Apr. 2003
 - *Int'l Symp. on Low Power Electronics and Design*, Monterey, CA, Aug. 1997
 - *Int'l Symp. on Low Power Design*, Laguna, CA, Apr. 1995
- Technical Chair
 - *Int'l Symp. on Physical Design*, Del Mar, CA, Apr. 2002
 - *Int'l Symp. on Low Power Electronics and Design*, Monterey, CA, Aug. 1996
 - *Int'l Workshop on Low Power Design*, Napa Valley, CA, Apr. 1994
- Executive Committee Member
 - *Int'l Symp. on Low Power Electronics and Design*, since 1996
 - *Int'l Conference on Computer Aided Design*, 2006-2010
- Technical Program Committee Member
 - *The 18th CSI International Symposium on Computer Architecture & Digital Systems*, Tehran, Iran, Oct. 7-8, 2015
 - *Int'l Conference on Hardware/Software Codesign and System Synthesis*, 2013-2015
 - *Symp. on Embedded Systems for Real-Time Multimedia (ESTIMedia)*, 2007-2012
 - *Symp. on VLSI Design, Automation and Test (VLSI-DAT)*, 2012
 - *2009 Great Lakes Symp. on VLSI*
 - *2008 IEEE Int'l Design and Test Workshop*, Tunisia
 - *Symp. on VLSI Design, Automation and Test (VLSI-DAT)*, Hsinchu, Taiwan, 2009
 - *NanoArch*, 2007-2008
 - *The 13th CSI Computer Conference (CSICC'08)*, Kish Island, Iran, March 9-11, 2007
 - *Conference on VLSI – Design Automation Technologies*, Taipei, Taiwan, 2006-2007
 - *1998-2000 Int'l Conference on Computer Aided Design*
 - *1998-2001 Int'l Symp. on Physical Design*
 - *1997-2000 Int'l Workshop on Logic Synthesis*
 - *2000 Asia and South Pacific Design Automation Conference*
 - *1998-2003 Design Automation and Test in Europe Conference*
 - *1994-2000 Int'l Symp. on Low Power Electronics and Design*
 - *2001-2003 Int'l Symp. on Quality of Electronic Design*

US Patents

- [S1] Nazemi; Mahdi (Los Angeles CA); Fayyazi; Arash (Los Angeles CA); Esmaili; Amirehossein (Los Angeles CA); Pedram; Massoud (Beverly Hills CA), "System and Method for Hybrid Arithmetic and Logic Processing of Neural Networks," United States Patent filed on Dec. 22, 2022. EFS ID: 47261286. Application Number: 20230281432, Publication Date: Sep 7, 2023.
- [S2] Zhang; Bo (Los Angeles CA); Chen; Zeming (Los Angeles CA); Pedram; Massoud (Beverly Hills CA); "Design of High-Performance and Scalable Montgomery Modular Multiplier Circuits," US Patent filed on Apr 23, 2021, US Application Publication number: 20230185537, US Application Publication Date: Jun 15, 2023, International Patent filed on May 7, 2021, International Application Number: PCT/US2021/028896.
- [S3] Pasandi; Ghasem (Los Angeles, CA); Pedram; Massoud (Beverly Hills CA), "Efficient pipelined architecture for superconducting single flux quantum logic circuits utilizing dual clocks," United States Patent **11,303,281**, Issued: Apr. 4, 2022, Filed: Apr. 5, 2021.
- [S4] Katam; Naveen (Los Angeles, CA); Mukhanov; Oleg (Elmsford, NY); Pedram; Massoud (Los Angeles, CA), "Superconductive magnetic field programmable gate array," United States Patent **10,707,873**, Issued: Jul. 7, 2020, Filed: Mar. 21, 2019.
- [S5] Pedram; Massoud (Beverly Hills, CA); Iranli; Ali (Los Angeles, CA), "Dynamic backlight scaling for power minimization in a backlit TFT-LCD," United States Patent **8,094,118**, Issued: Jan 10, 2012, Filed: Mar 2, 2006.
- [S6] Fallah; Farzan (San Jose, CA); Pakbaznia; Ehsan (Los Angeles, CA); Pedram; Massoud (Beverly Hills, CA); "Sizing and placement of charge recycling (CR) transistors in multithreshold complementary metal-oxide-semiconductor (MTCMOS) circuit," United States Patent **7,834,684**, Issued: Nov 16, 2010, Filed: Oct 31, 2008.
- [S7] Fallah; Farzan (San Jose, CA); Abdollahi; Afshin (Los Angeles, CA); Pedram; Massoud (Los Angeles, CA); "Power mode transition in multi-threshold complementary metal oxide semiconductor (MTCMOS) circuits," United States Patent **7,613,942**, Issued: November 3, 2009, Filed: May 31, 2006.
- [S8] Fallah; Farzan (San Jose, CA); Amelifard; Behnam (Los Angeles, CA); Pedram; Massoud (Beverly Hills, CA); "Setting threshold voltages of cells in a memory block to reduce leakage in the memory block," United States Patent **7,573,775**, Issued: August 11, 2009, Filed: February 9, 2007.
- [S9] Fallah; Farzan (San Jose, CA); Amelifard; Behnam (Los Angeles, CA); Pedram; Massoud (Beverly Hills, CA); "PG-Gated Data Retention Technique for Reducing Leakage in Memory Cells," United States Patent **7,447,101**, Issued: November 4, 2008, Filed: December 22, 2006.
- [S10] Fallah; Farzan (San Jose, CA); Pakbaznia; Ehsan (Los Angeles, CA); Pedram; Massoud (Beverly Hills, CA); "Recycling Charge to Reduce Energy Consumption During Mode Transition in Multi-Threshold Complementary Metal-Oxide-Semiconductor (MTCMOS) Circuits," United States Patent **7,400,175**, Issued: July 15, 2008, Filed: May 30, 2007.
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